1 A model with passive investors

1.1 The model

Consider a version of the problem with only two investors, $j = \{0, 1\}$, where the type-0 represent passive investors, who maintains a fixed portfolio share, and type-1 represents an active investor, who continuous rebalance their portfolio.

Financial markets. Investors have access to a riskless asset, which pays instantaneous return r_t, and a risky asset, which is a claim to dividends Y_t that follows the process:

$$\frac{dY_t}{Y_t} = \mu dt + \sigma dZ_t. \tag{1}$$

The price of the risky asset is given by P_t, which follows the process:

$$\frac{dP_t}{P_t} = \mu_{P,t} dt + \sigma_{P,t} dZ_t. \tag{2}$$

Let $p_t \equiv \frac{P_t}{Y_t}$ denote the price-dividend ratio, so the risk premium π_t and return volatility are given by:

$$\pi_t = \frac{1}{p_t} + \mu + \mu_{p,t} + \sigma\sigma_{p,t} - r_t, \qquad \sigma_{R,t} = \sigma + \sigma_{p,t} \tag{3}$$

The drift and diffusion of the price-dividend ratio are given by Ito's lemma:

$$\mu_{p,t} = \frac{p_{x,t}}{p_t}\mu_{x,t} + \frac{1}{2}\frac{p_{xx,t}}{p_t}\sigma_{x,t}^2, \qquad \sigma_{p,t} = \frac{p_{x,t}}{p_t}\sigma_{x,t}. \tag{4}$$

In terms of the dividend yield, $y_t \equiv \frac{Y_t}{P_t}$, we obtain

$$\pi_t = y_t + \mu - \mu_{y,t} + \sigma_{y,t}^2 - \sigma\sigma_{y,t} - r_t, \qquad \sigma_{R,t} = \sigma - \sigma_{y,t}. \tag{5}$$

Investors. The portfolio share of passive investors is exogenously given by $\alpha_{0,t} = \overline{\alpha}_p$, and the portfolio share of the active investor is given by

$$\alpha_{1,t} = \frac{\pi_t}{\gamma_1 \sigma_{R,t}^2} + \varsigma_{1,t}, \tag{6}$$

where $\varsigma_{1,t} \equiv \frac{1-\gamma_1^{-1}}{\psi-1}\frac{\sigma_{c_1,t}}{\sigma_{R,t}}$.

The consumption-wealth ratio of a type-j investor is given by

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \pi_t\alpha_{j,t} - \frac{\gamma_j}{2}\sigma_{R,t}^2\alpha_{j,t}^2\right] + \xi_{j,t}, \tag{7}$$

where

$$\xi_{j,t} \equiv \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{R,t}\alpha_{j,t} + \frac{\psi-\gamma_j}{1-\psi}\frac{\sigma_{c,t}^2}{2}. \tag{8}$$

Market clearing. The market clearing condition for risky assets and goods are given by

$$(1-x)\overline{\alpha}_p + x\alpha_{1,t} = 1, \qquad (1-x)c_{0,t} + xc_{1,t} = \frac{1}{p_t}. \tag{9}$$

Rearranging the expressions above, we obtain

$$\alpha_1(x_t) = \frac{1-(1-x)\overline{\alpha}_p}{x}. \tag{10}$$

The risk premium is given by

$$\pi_t = \gamma_1 \sigma_{R,t}^2 \left[\alpha_1(x_t) - \varsigma_{1,t}\right]. \tag{11}$$

State variable dynamics. The aggregate state variables corresponds to the share of wealth of active investors x, which evolves according to:

$$dx_t = x_t \left[(\pi_t - \sigma_{R,t}^2)(\alpha_{1,t}-1) + \kappa \frac{\omega_1 - x_t}{x_t}\right] dt + x_t(\alpha_{1,t}-1)\sigma_{R,t} dZ_t. \tag{12}$$

Notice that we can write $\sigma_{x,t}$ as follows:

$$\sigma_{x,t} = x_t(\alpha_{1,t}-1)\left(\sigma + \frac{p_{x,t}}{p_t}\sigma_{x,t}\right) \Rightarrow \sigma_{x,t} = \frac{x_t(\alpha_{1,t}-1)}{1 - x_t(\alpha_{1,t}-1)\frac{p_{x,t}}{p}}\sigma. \tag{13}$$

The differential equation.

$$c_{j,t} = \psi\rho + (\psi^{-1}-1)\left[y_t + \mu - \mu_{y,t} + \sigma_{y,t}^2 - \sigma\sigma_{y,t} + \pi_t(\alpha_{j,t}-1) - \frac{\gamma_j}{2}\sigma_{R,t}^2\alpha_{j,t}^2\right] + \xi_{j,t}. \tag{14}$$

1.2 A convenient change of variables

Define $s_t \equiv -\log x_t$, so $x_t = e^{-s_t}$. Let $w_j(s_t) \equiv \log c_j(e^{-s_t})$ and $w(s_t) \equiv \log y(e^{-s_t})$. The derivatives of $w_{j,t}$ are given by:

$$w_{j,s} = -\frac{c_{j,x}}{c_j}x, \qquad w_{j,ss} = x^2\frac{c_{j,xx}}{c_j} - (w_{j,s} + w_{j,s}^2). \tag{15}$$

and the derivative of $w(s_t)$ is given by:

$$w_s = -\frac{y_x}{y}x = x_t\frac{c_2 - c_1}{y} + \frac{x_t c_1}{y}w_{1,s} + \frac{(1-x_t)c_2}{y}w_{2,s}. \tag{16}$$

The diffusion of x_t is given by:

$$\sigma_{x,t} = \frac{(1-x_t)(1-\alpha_p)\sigma}{1 + (1-x_t)(1-\alpha_p)\frac{y_{x,t}}{y_t}} = \frac{x_t(1-x_t)(1-\alpha_p)\sigma}{x_t - (1-x_t)(1-\alpha_p)w_{s,t}} \tag{17}$$

The return volatility is given by:

$$\sigma_{R,t} = \sigma + \frac{w_{s,t}}{x_t}\sigma_{x,t} = \frac{x_t\sigma}{x_t - (1-x_t)(1-\alpha_p)w_{s,t}}. \tag{18}$$

The diffusion of $c_{j,t}$ is given by:

$$\sigma_{c_j,t} = \frac{c_{j,x,t}}{c_{j,t}} x_t(\alpha_{j,t}-1)\sigma = -w_{j,s,t}(\alpha_{j,t}-1)\sigma, \tag{19}$$

The scaled hedging demand is given by:

$$x_t\varsigma_{j,t} = \frac{1-\gamma_j^{-1}}{\psi-1} x_t \frac{c_{j,x}}{c_j}\frac{\sigma_{x,t}}{\sigma_{R,t}} = -\frac{1-\gamma_j^{-1}}{\psi-1} w_{j,s,t}(1-x_t)(1-\alpha_p). \tag{20}$$

Define the scaled risk premium as $\tilde{\pi}_t \equiv \pi_t x_t$, which can be expressed as follows:

$$\tilde{\pi}_t = \gamma_1\sigma_{R,t}^2\left[x_t + x_t(\alpha_{1,t}-1)\left(1 + w_{j,s,t}\frac{1-\gamma_1^{-1}}{\psi-1}\frac{\sigma}{\sigma_{R,t}}\right)\right]. \tag{21}$$

Using the fact that $x_t(\alpha_{1,t}-1) = (1-x_t)(1-\alpha_p)$, we obtain

$$\tilde{\pi}_t = \gamma_1\sigma_{R,t}^2\left[x_t + (1-x_t)(1-\alpha_p)\left(1 + w_{j,s,t}\frac{1-\gamma_1^{-1}}{\psi-1}\right)\right]. \tag{22}$$

2 A heterogeneous-agents model

2.1 The model

Consider a version of the problem with two investors, $j = \{1, 2\}$, which differ in their level of risk aversion, $\gamma_1 \neq \gamma_2$.

Financial markets. Investors have access to a riskless asset, which pays instantaneous return r_t, and a risky asset, which is a claim to dividends Y_t that follows the process:

$$\frac{dY_t}{Y_t} = \mu dt + \sigma dZ_t. \tag{23}$$

The price of the risky asset is given by P_t, which follows the process:

$$\frac{dP_t}{P_t} = \mu_{P,t}dt + \sigma_{P,t}dZ_t. \tag{24}$$

Let $p_t \equiv \frac{P_t}{Y_t}$ denote the price-dividend ratio, so the risk premium π_t and return volatility are given by:

$$\pi_t = \frac{1}{p_t} + \mu + \mu_{p,t} + \sigma\sigma_{p,t} - r_t, \qquad \sigma_{R,t} = \sigma + \sigma_{p,t} \tag{25}$$

The drift and diffusion of the price-dividend ratio are given by Ito's lemma:

$$\mu_{p,t} = \frac{p_{x,t}}{p_t}\mu_{x,t} + \frac{1}{2}\frac{p_{xx,t}}{p_t}\sigma_{x,t}^2, \qquad \sigma_{p,t} = \frac{p_{x,t}}{p_t}\sigma_{x,t}. \tag{26}$$

Investors. The portfolio share of investor j is given by

$$\alpha_{j,t} = \frac{\pi_t}{\gamma_j \sigma_{R,t}^2} + \varsigma_{j,t}, \tag{27}$$

where $\varsigma_{j,t} \equiv \frac{1-\gamma_j^{-1}}{\psi-1}\frac{\sigma_{c_j,t}}{\sigma_{R,t}}$.

The consumption-wealth ratio is given by

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \pi_t\alpha_{j,t} - \frac{\gamma_j}{2}\sigma_{R,t}^2\alpha_{j,t}^2\right] + \xi_{j,t}, \tag{28}$$

where

$$\xi_{j,t} \equiv \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{R,t}\alpha_{j,t} + \frac{\psi-\gamma_j}{1-\psi}\frac{\sigma_{c_j,t}^2}{2}. \tag{29}$$

Market clearing. The market clearing condition for risky assets and goods are given by

$$\sum_{j=1}^{2} x_{j,t}\alpha_{j,t} = 1, \qquad \sum_{j=1}^{2} x_{j,t}c_{j,t} = \frac{1}{p_t}, \tag{30}$$

where $x_{j,t}$ is the wealth share of investor j.

The risk premium is given by

$$\pi_t = \gamma_t\sigma_{R,t}^2\left[1 - \sum_{j=1}^{2} x_{j,t}\varsigma_{j,t}\right], \tag{31}$$

where $\gamma_t \equiv \left[\sum_{j=1}^{2} x_{j,t}\gamma_j^{-1}\right]^{-1}$.

The interest rate can be computed as follows:

$$r_t = \frac{1}{p_t} + \mu + \mu_{p,t} + \sigma\sigma_{p,t} - \pi_t. \tag{32}$$

State variable dynamics. Let $x_t \equiv x_{1,t}$, so the aggregate state variables corresponds to the wealth share of type-1 investors, which evolves according to:

$$dx_t = x_t\left[(\pi_t - \sigma_{R,t}^2)(\alpha_{1,t}-1) + \kappa\frac{\omega_1 - x_t}{x_t}\right]dt + x_t(\alpha_{1,t}-1)\sigma_{R,t}dZ_t. \tag{33}$$

Given the expression for the risk premium, we can write the portfolio share as follows:

$$\alpha_{j,t} = \frac{\gamma_t}{\gamma_j}\left[1 - \sum_{j=1}^{2} x_{j,t}\varsigma_{j,t}\right] + \varsigma_{j,t}. \tag{34}$$

Multiplying by $\sigma_{R,t}$, and expanding, we obtain:

$$\alpha_{j,t}\sigma_{R,t} = \frac{\gamma_t}{\gamma_j}\left[\sigma - \frac{y_x}{y}\sigma_{x,t} - \sum_{k=1}^{2} x_{k,t}\frac{1-\gamma_k^{-1}}{\psi-1}\frac{c_{k,x,t}}{c_{k,t}}\sigma_{x,t}\right] + \frac{1-\gamma_j^{-1}}{\psi-1}\frac{c_{j,x,t}}{c_{j,t}}\sigma_{x,t}. \tag{35}$$

Taking the between the two agents, we obtain

$$\left(\alpha_{1,t} - \alpha_{2,t}\right)\sigma_{R,t} = \left(\frac{\gamma_t}{\gamma_1} - \frac{\gamma_t}{\gamma_2}\right)\sigma + \chi_{\Delta\alpha,x}(x_t)\sigma_{x,t}, \tag{36}$$

where

$$\chi_{\Delta\alpha,x}(x_t) \equiv -\left(\frac{\gamma_t}{\gamma_1} - \frac{\gamma_t}{\gamma_2}\right)\left[\frac{y_x}{y} + \sum_{k=1}^{2} x_{k,t}\frac{1-\gamma_k^{-1}}{\psi-1}\frac{c_{k,x,t}}{c_{k,t}}\right] + \frac{1-\gamma_1^{-1}}{\psi-1}\frac{c_{1,x,t}}{c_{1,t}} - \frac{1-\gamma_2^{-1}}{\psi-1}\frac{c_{2,x,t}}{c_{2,t}}. \tag{37}$$

Notice that we can write $\sigma_{x,t}$ as follows:

$$\sigma_{x,t} = x_t(1-x_t)(\alpha_{1,t} - \alpha_{2,t})\sigma_{R,t} \Rightarrow \sigma_{x,t} = \frac{x_t(1-x_t)\left(\frac{\gamma_t}{\gamma_1} - \frac{\gamma_t}{\gamma_2}\right)}{1 - x_t(1-x_t)\chi_{\Delta\alpha,x}(x_t)}\sigma. \tag{38}$$

Dividend yield. The dividend yield is given by $y_t \equiv \frac{1}{p_t}$, so the law of motion of y_t is given by

$$\frac{dy_t}{y_t} = -\frac{dp_t}{p_t} + \left(\frac{dp_t}{p_t}\right)^2 = \left[-\mu_{p,t} + \sigma_{p,t}^2\right]dt - \sigma_{p,t}dZ_t. \tag{39}$$

The dividend yield is given by

$$y_t = x_t c_{1,t} + (1-x_t)c_{2,t}. \tag{40}$$

The derivative with respect to x_t is given by

$$\frac{y_{x,t}}{y_t} = \frac{c_{1,t} - c_{2,t}}{y_t} + \frac{x_t c_{1,t}}{y_t}\frac{c_{1,x,t}}{c_{1,t}} + \frac{(1-x_t)c_{2,t}}{y_t}\frac{c_{2,x,t}}{c_{2,t}}. \tag{41}$$

The second derivative is given by

$$\frac{y_{xx,t}}{y_t} = 2\frac{c_{1,x,t} - c_{2,x,t}}{y_t} + \frac{x_t c_{1,t}}{y_t}\frac{c_{1,xx,t}}{c_{1,t}} + \frac{(1-x_t)c_{2,t}}{y_t}\frac{c_{2,xx,t}}{c_{2,t}}. \tag{42}$$

The system of differential equations. We need to solve for $(c_1(x_t), c_2(x_t))$. The system of differential equations for $c_{1,t}$ and $c_{2,t}$ is given by:

$$c_{j,t} = \psi\rho + (1-\psi)\left[x_t c_{1,t} + (1-x_t)c_{2,t} + \mu - \mu_{y,t} + \sigma_{y,t}^2 - \sigma\sigma_{y,t} + \pi_t(\alpha_{j,t} - 1) - \frac{\gamma_j}{2}\sigma_{R,t}^2\alpha_{j,t}^2\right] + \xi_{j,t}, \tag{43}$$

where $y_t = x_t c_{1,t} + (1 - x_t) c_{2,t}$ is given by

$$y_t = \rho + (\psi^{-1} - 1) \left[\mu - \mu_{y,t} + \sigma_{y,t}^2 - \sigma \sigma_{y,t} - \sum_{j=1}^{2} x_{j,t} \frac{\gamma_j}{2} \sigma_{R,t}^2 \alpha_{j,t}^2 \right] + \psi^{-1} \xi_t, \tag{44}$$

2.2 Regular perturbation

Suppose $\gamma_j = \gamma(1 + \hat{\gamma}_j \epsilon)$ and $\kappa = \hat{\kappa}\epsilon$, where ϵ is a small parameter. We can then write the the consumption-wealth ratios as follows:

$$c_j(x; \epsilon) = \sum_{k=0}^{\infty} c_{j,k}(x) \epsilon^k. \tag{45}$$

Zeroth-order term. The zeroth-order term is given by

$$c_{j,0}(x) = \rho + \left(\psi^{-1} - 1\right) \left[\mu - \frac{\gamma \sigma^2}{2} \right]. \tag{46}$$

The risk-premium and interest rate are given by

$$\pi_0(x) = \gamma \sigma^2, \qquad r_0(x) = y^\star + \mu - \gamma \sigma^2, \tag{47}$$

where $y^\star \equiv c_{j,0}(x)$. The portfolio share is given by $\alpha_{j,0}(x) = 1$.

First-order correction. Notice that $\mu_{c_j}(x, \epsilon) = \mathcal{O}(\epsilon^2)$ and $\sigma_{c_j}(x, \epsilon) = \mathcal{O}(\epsilon^2)$, so the term $\xi_{j,1}(x, \epsilon) = \mathcal{O}(\epsilon^2)$. This implies that $\sigma_y(x; \epsilon) = \mathcal{O}(\epsilon^2)$. We can then write the risk premium as follows:

$$\pi_1(x) = \gamma \sigma^2 \sum_{j=1}^{2} x_{j,t} \hat{\gamma}_j. \tag{48}$$

The portfolio share is then given by

$$\alpha_{j,1}(x) = \frac{\pi_1(x)}{\gamma \sigma^2} - \hat{\gamma}_j = -(1 - x_j)(\hat{\gamma}_j - \hat{\gamma}_{-j}). \tag{49}$$

The dividend yield is then given by

$$y_1(x) = (1 - \psi^{-1}) \frac{\gamma \sigma^2}{2} \sum_{j=1}^{2} x_{j,t} \hat{\gamma}_j. \tag{50}$$

The consumption-wealth ratio is then given by

$$c_{j,1}(x) = (1 - \psi) \frac{\gamma \sigma^2}{2} \left[\left(1 - \psi^{-1}\right) \sum_{k=1}^{2} x_{k,t} \hat{\gamma}_k - \hat{\gamma}_j \right]. \tag{51}$$

The diffusion term of x is given by:

$$\sigma_{x,1}(x) = x\alpha_{1,1}(x)\sigma = x(1-x)\left[\hat{\gamma}_2 - \hat{\gamma}_1\right]\sigma. \tag{52}$$

The drift term of x is given by:

$$\mu_{x,1}(x) = x(1-x)(\gamma-1)\sigma^2(\hat{\gamma}_2 - \hat{\gamma}_1) + \hat{\kappa}(\omega_1 - x). \tag{53}$$

Second-order correction. The diffusion term of y is given by:

$$\sigma_{y,2}(x) = \frac{y_1(x)}{y_0(x)}\sigma_{x,1}(x) = -(1-\psi^{-1})\frac{\gamma\sigma^2}{2y^\star}\left[\hat{\gamma}_2 - \hat{\gamma}_1\right]^2\sigma. \tag{54}$$

The drift term of y is given by:

$$\mu_{y,2}(x) = -(1-\psi^{-1})\frac{\gamma\sigma^2}{2y^\star}\left[\hat{\gamma}_2 - \hat{\gamma}_1\right]\left[x(1-x)(\gamma-1)\sigma^2(\hat{\gamma}_2 - \hat{\gamma}_1) + \hat{\kappa}(\omega_1 - x)\right]. \tag{55}$$

Notice that $\mu_{c_j,2}(x)$ and $\sigma_{c_j,2}(x)$ are given by:

$$\mu_{c_j,2}(x) = -(\psi-1)\mu_{y,2}(x), \qquad \sigma_{c_j,2}(x) = -(\psi-1)\sigma_{y,2}(x). \tag{56}$$

The hedging demand is given by:

$$\varsigma_{j,2}(x) = -(1-\gamma^{-1})\frac{\sigma_{y,2}(x)}{\sigma}. \tag{57}$$

The average risk aversion can be expanded as follows:

$$\gamma_m(x;\epsilon) = \gamma + \gamma_{m,1}(x)\epsilon + \gamma_{m,2}(x)\epsilon^2 + \mathcal{O}(\epsilon^3), \tag{58}$$

where

$$\gamma_{m,1}(x) \equiv \gamma\sum_{j=1}^{2} x_j\hat{\gamma}_j, \quad \text{and} \quad \gamma_{m,2}(x) \equiv -\gamma\left(\sum_{j=1}^{2} x_j\hat{\gamma}_j^2 - \left(\sum_{j=1}^{2} x_j\hat{\gamma}_j\right)^2\right).$$

The risk premium is given by:

$$\pi_2(x) = \gamma\sigma^2\left[\frac{\gamma_{m,2}(x)}{\gamma} - (1+\gamma^{-1})\frac{\sigma_{y,2}(x)}{\sigma}\right]. \tag{59}$$

The portfolio share is then given by:

$$\alpha_{j,2}(x) = \frac{\gamma_{m,2}(x)}{\gamma} + \hat{\gamma}_j^2 - \hat{\gamma}_j\frac{\pi_1(x)}{\pi_0(x)}. \tag{60}$$

The shifter of the consumption-wealth ratio is given by:

$$\xi_{j,2}(x) = -(\psi - 1)\left[\mu_{y,2}(x) + (1-\gamma)\sigma_{y,2}(x)\sigma\right]. \tag{61}$$

The dividend yield is then given by:

$$y_2(x) = (1-\psi^{-1})\gamma\sigma^2\left[\sum_{j=1}^{2} x_{j,t}\hat{\gamma}_j\alpha_{j,1}(x) + \sum_{j=1}^{2} x_{j,t}\frac{\alpha_{j,1}^2(x)}{2}\right]. \tag{62}$$

The consumption-wealth ratio is then given by:

$$c_{j,2}(x) = (1-\psi)\left[y_2(x) + \pi_1(x)\alpha_{j,1}(x) - \gamma\sigma^2\hat{\gamma}_j\alpha_{j,1}(x) - \frac{\gamma\sigma^2}{2}\alpha_{j,1}^2(x)\right]. \tag{63}$$

The diffusion term of x is given by:

$$\sigma_{x,2}(x) = x\alpha_{1,2}(x)\sigma. \tag{64}$$

The drift term of x is given by:

$$\mu_{x,2}(x) = x\left[\pi_1(x)\alpha_{1,1}(x) + (\gamma-1)\sigma^2\alpha_{1,2}(x)\right]. \tag{65}$$

Higher-order perturbation Consider the expression for the consumption-wealth ratio:

$$\begin{bmatrix} c_{1,t} \\ c_{2,t} \end{bmatrix} = \begin{bmatrix} 1-(1-\psi)x & -(1-\psi)(1-x) \\ -(1-\psi)x & 1-(1-\psi)(1-x) \end{bmatrix}^{-1} \begin{bmatrix} b_{1,t} \\ b_{2,t} \end{bmatrix}, \tag{66}$$

where

$$b_{j,t} = \psi\rho + (1-\psi)\left[\mu - \mu_{y,t} + \sigma_{y,t}^2 - \sigma\sigma_{y,t} + \pi_t(\alpha_{j,t}-1) - \frac{\gamma_j}{2}\sigma_{R,t}^2\alpha_{j,t}^2\right] + \xi_{j,t}, \tag{67}$$

The k-th order correction depends then on $b_{j,k}(x)$, which depends only on $c_{j,l}(x)$ for $l < k$. Hence, we can use the expression above to compute the higher-order perturbation.

2.3 Inner solution

Consider the solution in the inner region, where $x \to 0$. Let's work with the transformed variable: $\hat{x} = x/\epsilon$, so the consumption-wealth ratio is given by

$$c_j(x;\epsilon) = \hat{c}_j(\hat{x};\epsilon), \qquad \hat{x} \equiv \frac{x}{\epsilon}, \tag{68}$$

which gives the derivatives

$$c_{j,x}(x) = \epsilon^{-1}\hat{c}_{j,\hat{x}}(\hat{x}), \qquad c_{j,xx}(x) = \epsilon^{-2}\hat{c}_{j,\hat{x}\hat{x}}(\hat{x}). \tag{69}$$

We are interested in computing the expansion:

$$\hat{c}_j(\hat{x};\epsilon) = \hat{c}_{j,0}(\hat{x}) + \hat{c}_{j,1}(\hat{x})\epsilon + \hat{c}_{j,2}(\hat{x})\epsilon^2 + \dots. \tag{70}$$

The drift of the state variable is given by

$$\sigma_{\hat{x},t} = \frac{\hat{x}_t(\alpha_{1,t}-1)}{1+\hat{x}_t(\alpha_{1,t}-1)\frac{\hat{y}_{\hat{x},t}}{\hat{y}}}\sigma = \mathcal{O}(\epsilon), \tag{71}$$

as $\alpha_{1,t} - 1 = \mathcal{O}(\epsilon)$.

This implies that the diffusion of y is given by

$$\sigma_y(x;\epsilon) = \frac{\hat{y}_{\hat{x}}(\hat{x};\epsilon)}{\hat{y}(\hat{x};\epsilon)}\sigma_{\hat{x}}(\hat{x};\epsilon) = \mathcal{O}(\epsilon^2), \tag{72}$$

as $\hat{y}_{\hat{x}}(\hat{x};\epsilon) = \mathcal{O}(\epsilon)$.

Similarly, the drift of the state variable is given by

$$\mu_{\hat{x}}(\hat{x};\epsilon) = \hat{x}_t\left[(\pi_t - \sigma_{R,t}^2)(\alpha_{1,t}-1) + \hat{\kappa}\frac{\omega_1 - \hat{x}_t\epsilon}{\hat{x}_t}\right] = \mathcal{O}(1), \tag{73}$$

as $\alpha_{1,t} - 1 = \mathcal{O}(\epsilon)$ and $\hat{\kappa} = \mathcal{O}(1)$.

The drift of y is given by

$$\mu_y(x;\epsilon) = \frac{\hat{y}_{\hat{x}}(\hat{x};\epsilon)}{\hat{y}(\hat{x};\epsilon)}\mu_{\hat{x}}(\hat{x};\epsilon) + \frac{1}{2}\frac{\epsilon^{-2}\hat{y}_{\hat{x}\hat{x}}(\hat{x};\epsilon)}{\hat{y}(\hat{x};\epsilon)}(\sigma_{\hat{x}}(\hat{x};\epsilon)\epsilon)^2 \tag{74}$$

Notice that the second term is of order $\mathcal{O}(\epsilon^2)$, so $\mu_{\hat{y}}(\hat{x};\epsilon) = \mathcal{O}(\epsilon)$.

Zeroth-order term. The system of equations determining $\hat{c}_j(\hat{x};\epsilon)$ is given by:

$$\hat{c}_{j,t} = \psi\rho + (1-\psi)\left[\hat{y}_t + \mu - \mu_{y,t} + \sigma_{y,t}^2 - \sigma\sigma_{y,t} + \pi_t(\alpha_{j,t}-1) - \frac{\gamma_j}{2}\sigma_{R,t}^2\alpha_{j,t}^2\right] + \xi_{j,t}, \tag{75}$$

The zero-order term of the inner expansion of the consumption-wealth is given by

$$\hat{c}_{j,0}(\hat{x}) = \psi\rho + (1-\psi)\left[\hat{y}_0(x) + \mu - \mu_{y,0} - \frac{\gamma}{2}\sigma^2\right] + \mu_{y,0}. \tag{76}$$

Using the fact that $\hat{c}_{j,0} = \hat{y}_0$, we obtain

$$\hat{y}_0 = \rho + (\psi^{-1}-1)\left[\mu - \frac{\gamma\sigma^2}{2}\right] = y^\star. \tag{77}$$

The risk premium and interest rate are given by

$$\hat{\pi}_0(\hat{x}) = \gamma\sigma^2, \qquad \hat{r}_0(\hat{x}) = \rho + \psi^{-1}\mu - \frac{\gamma\sigma^2}{2}(1+\psi^{-1}). \tag{78}$$

First-order term. The risk premium is given by

$$\hat{\pi}_t = \hat{\gamma}_m(\hat{x};\epsilon)\hat{\sigma}_R(\hat{x};\epsilon)^2 \left[1 - \sum_{j=1}^{2} x_j \hat{\varsigma}_j(\hat{x};\epsilon)\right], \tag{79}$$

where

$$\hat{\gamma}_m(\hat{x};\epsilon) = \frac{\gamma}{\frac{\hat{x}\epsilon}{1+\hat{\gamma}_1\epsilon} + \frac{1-\hat{x}\epsilon}{1+\hat{\gamma}_2\epsilon}} = \frac{\gamma}{1 - \left(\frac{\hat{x}+\hat{\gamma}_2}{1+\hat{\gamma}_2\epsilon} - \frac{\hat{x}}{1+\hat{\gamma}_1\epsilon}\right)\epsilon} = \gamma\left[1 + \hat{\gamma}_2\epsilon\right] + \mathcal{O}(\epsilon^2). \tag{80}$$

The first-order correction for the risk premium is given by

$$\hat{\pi}_1(\hat{x}) = \gamma\sigma^2\hat{\gamma}_2. \tag{81}$$

The portfolio share of type j is given by

$$\hat{\alpha}_{j,1}(\hat{x}) = \hat{\gamma}_2 - \hat{\gamma}_j. \tag{82}$$

The dividend yield is given by

$$\hat{y}(\hat{x};\epsilon) = \hat{x}\epsilon\hat{c}_1(\hat{x};\epsilon) + (1-\hat{x}\epsilon)\hat{c}_2(\hat{x};\epsilon) = y^\star + \hat{c}_{2,1}(\hat{x})\epsilon + \mathcal{O}(\epsilon^2). \tag{83}$$

Hence, $\hat{y}_1(\hat{x}) = \hat{c}_{2,1}(\hat{x})$ and $\mu_{\hat{y},1} = \mu_{\hat{c}_2,1} = \frac{\hat{y}_{\hat{x}}(\hat{x})}{y^\star}\hat{\kappa}\omega_1$.

From the expression for the consumption-wealth ratio of the type 2 agent is:

$$\hat{y}_{1,\hat{x}}(\hat{x}) - \delta\hat{y}_1 = -(1-\psi^{-1})\delta\frac{\gamma\sigma^2}{2}\hat{\gamma}_2, \tag{84}$$

where $\delta \equiv \frac{y^\star}{\hat{\kappa}\omega_1}$.

The solution to this differential equation is given by

$$e^{-\delta\hat{x}}\hat{y}_1(\hat{x}) - \hat{y}_1(0) = -(1-\psi^{-1})\delta\frac{\gamma\sigma^2}{2}\hat{\gamma}_2\frac{1-e^{-\delta\hat{x}}}{\delta}. \tag{85}$$

As $\delta > 0$ and the solution is bounded, we obtain

$$\hat{y}_1(\hat{x}) = (1-\psi^{-1})\frac{\gamma\sigma^2}{2}\hat{\gamma}_2. \tag{86}$$

This implies that the drift of $\hat{y}$ is $\mu_{\hat{y},1}(\hat{x}) = 0$.

The interest rate is given by

$$\hat{r}_1(\hat{x}) = \hat{y}_1(\hat{x}) - \hat{\pi}_1(\hat{x}). \tag{87}$$

The consumption-wealth ratio of the type-1 investor is:

$$\hat{c}_{1,1}(\hat{x}) = (1-\psi)\left[\hat{y}_1(\hat{x}) - \frac{\gamma\sigma^2}{2}\hat{\gamma}_1\right] + \frac{\hat{c}_{1,1,x}(\hat{x})}{y^\star}\hat{\kappa}\omega_1. \tag{88}$$

Solving the differential equation, we obtain

$$e^{-\delta\hat{x}}\hat{c}_{1,1}(\hat{x}) - \hat{c}_{1,1}(0) = -(1-\psi)\delta\left[\hat{y}_1(\hat{x}) - \frac{\gamma\sigma^2}{2}\hat{\gamma}_1\right]\frac{1-e^{-\delta\hat{x}}}{\delta}. \tag{89}$$

This implies that

$$\hat{c}_{1,1}(\hat{x}) = (1-\psi)\left[\hat{y}_1(\hat{x}) - \frac{\gamma\sigma^2}{2}\hat{\gamma}_1\right] = \hat{y}_1(\hat{x}) - (\psi-1)\frac{\gamma\sigma^2}{2}\left[\hat{\gamma}_2 - \hat{\gamma}_1\right]. \tag{90}$$

The drift of $\hat{x}$ is given by

$$\sigma_{\hat{x},1}(\hat{x}) = \hat{x}(\hat{\gamma}_2 - \hat{\gamma}_1)\sigma. \tag{91}$$

The drift of $\hat{x}$ is given by

$$\mu_{\hat{x},1}(\hat{x}) = \hat{x}(\gamma-1)\sigma^2(\hat{\gamma}_2 - \hat{\gamma}_1) - \hat{\kappa}\hat{x}. \tag{92}$$

Second-order term. Because the first-order correction of $\hat{c}_j$ is constant, we have that $\sigma_{\hat{c}_j,2} = \sigma_{y,2} = 0$. This implies that the hedging demands are also equal to zero: $\hat{\varsigma}_{j,2}(\hat{x}) = 0$.

The second-order correction for the average risk aversion is

$$\hat{\gamma}_{m,2}(\hat{x}) = -\gamma(\hat{\gamma}_2 - \hat{\gamma}_1)\hat{x}. \tag{93}$$

The risk premium is given by

$$\hat{\pi}_2(\hat{x}) = -\gamma\sigma^2(\hat{\gamma}_2 - \hat{\gamma}_1)\hat{x}. \tag{94}$$

The portfolio share of type j is given by:

$$\hat{\alpha}_{j,2}(\hat{x}) = \frac{\hat{\pi}_2(\hat{x})}{\gamma\sigma^2} - \frac{\hat{\pi}_1(\hat{x})}{\gamma\sigma^2}\hat{\gamma}_j + \hat{\gamma}_j^2. \tag{95}$$

The drift of $\hat{y}$ is given by

$$\mu_{\hat{y},2}(\hat{x}) = \frac{\hat{y}_{2,\hat{x}}(\hat{x})}{y^\star}\hat{\kappa}\omega_1. \tag{96}$$

The dividend yield is given by

$$\hat{y}_2(\hat{x}) = \hat{x}(\hat{c}_{1,1}(\hat{x}) - \hat{c}_{2,1}(\hat{x})) + \hat{c}_{2,2}(\hat{x}). \tag{97}$$

This implies that the drift of $\hat{y}$ satisfies the condition:

$$\mu_{\hat{y},2} = \frac{\hat{c}_{1,1}(\hat{x}) - \hat{c}_{2,1}(\hat{x})}{y^\star}\hat{\kappa}\omega_1 + \mu_{\hat{c}_2,2}. \tag{98}$$

The consumption-wealth ratio for $j = 2$ satisfies the condition:

$$\hat{c}_{j,2}(\hat{x}) = (1-\psi)\left[\hat{y}_2(\hat{x}) - \mu_{\hat{y},2}(\hat{x}) + (\hat{\pi}_1(\hat{x}) - \hat{\gamma}_j\gamma\sigma^2)\hat{\alpha}_{j,1}(\hat{x}) - \frac{\gamma}{2}\sigma^2\hat{\alpha}_{j,1}(\hat{x})^2\right] + \mu_{\hat{c}_2,2}(\hat{x}) \tag{99}$$

For $j = 2$, we have $\hat{\alpha}_{j,1}(\hat{x}) = 0$. This implies that the consumption-wealth ratio for the type $j = 2$ satisfies:

$$\hat{c}_{2,2,\hat{x}}(\hat{x}) - \delta\hat{c}_{2,2}(\hat{x}) = -(\psi^{-1}-1)\left[\hat{x}\delta(\hat{c}_{1,1}(\hat{x}) - \hat{c}_{2,1}(\hat{x})) - (\hat{c}_{1,1}(\hat{x}) - \hat{c}_{2,1}(\hat{x}))\right] = -A(\delta\hat{x} - 1), \tag{100}$$

where $A \equiv (\psi^{-1}-1)\left[\hat{c}_{1,1}(\hat{x}) - \hat{c}_{2,1}(\hat{x})\right]$.

Solving the differential equation, we obtain

$$e^{-\delta\hat{x}}\hat{c}_{2,2}(\hat{x}) - \hat{c}_{2,2}(0) = A\hat{x}e^{-\delta\hat{x}} \Rightarrow \hat{c}_{2,2}(\hat{x}) = e^{\delta\hat{x}}\hat{c}_{2,2}(0) + A\hat{x} \tag{101}$$

To ensure the solution is bounded, we need to have that $\hat{c}_{2,2}(0) = 0$. This implies that the dividend yield is given by

$$\hat{y}_2(\hat{x}) = \psi^{-1}\left[\hat{c}_{1,1}(\hat{x}) - \hat{c}_{2,1}(\hat{x})\right]\hat{x}. \tag{102}$$

The consumption-wealth ratio for $j = 1$ is given by

$$\hat{c}_{j,2}(\hat{x}) = (1-\psi)\left[\hat{y}_2(\hat{x}) - \mu_{\hat{y},2}(\hat{x}) + (\hat{\pi}_1(\hat{x}) - \hat{\gamma}_j\gamma\sigma^2)\hat{\alpha}_{j,1}(\hat{x}) - \frac{\gamma}{2}\sigma^2\hat{\alpha}_{j,1}(\hat{x})^2\right] + \mu_{\hat{c}_2,2}(\hat{x}) \tag{103}$$

References